

At this presentation, we began by talking about Einstein and the early ideas that led to nuclear weapons, then connected that story to the Manhattan Project and Columbia University's role in it. We mentioned how the recent Oppenheimer movie has brought that history back into public conversation, and used it as a way to ask what those discoveries still mean for people today who live with the legacy of nuclear testing.

To give a sense of explosive power with the speaker, she first showed the Oklahoma City bombing, which used about 2.5 tons of TNT, with photos of the building before and after the attack. Then she switched to a slide on the Castle Bravo hydrogen bomb test in 1954, which had a yield of about 15 megatons of TNT. She explained that this was millions of times more powerful than the Oklahoma City bomb and that hydrogen bombs use fusion, the same process that powers the sun, rather than just fission. A world map of nuclear explosions since 1945 showed where tests have taken place, with many of the largest in the Pacific.

From there she focused on the Marshall Islands, showing a map with places like Bikini and Eniwetok circled and a historical list of nuclear tests carried out there. She explained that the Bravo test scattered radioactive fallout over nearby islands and communities. The talk then moved to more recent scientific work that measures the long term effects of that fallout. One slide from a PNAS paper showed gamma radiation levels on different islands, with histograms and maps that revealed hotspots. Another slide showed results from fieldwork on cesium 137 in fruit from the northern Marshall Islands, again from a PNAS study, and she explained how the team measured contamination directly on site.

She ended by talking about health impacts on people who were exposed to the Bravo fallout, saying that in the most affected population the average worked out to roughly two cancers per person over a lifetime. The final message was that the physics of nuclear weapons, the history of testing, and the ongoing medical and environmental consequences are all tightly connected, and that modern research, including work at Columbia, is still trying to understand and document that legacy.